Chapter 12: Linear Programming

School Worksheet: step-wise solutions

10 questions **27** marks **3** levels

Each row is one line of working. The red number in the right margin is what an examiner gives for that line, as in the CBSE marking scheme (½ is half a mark). Write every line that carries a mark.

Level 1

4 marks

1 MCQ — 1 mark

The corner points of the feasible region of an LPP are $(0, 2)$, $(3, 0)$, $(6, 0)$, $(6, 8)$ and $(0, 5)$. If $Z = 4x + 6y$, then the minimum value of Z occurs at:

(A) $(0, 2)$ only

(B) $(3, 0)$ only

(C) the mid-point of the line segment joining $(0, 2)$ and $(3, 0)$ only

(D) **any point on the line segment joining $(0, 2)$ and $(3, 0)$**

Formula / idea: If two adjacent corner points give the same optimal value, every point on the segment joining them is optimal.

1 Find $Z = 4x + 6y$ at each corner point.
$Z(0, 2) = 12, \quad Z(3, 0) = 12$
$Z(6, 0) = 24, \quad Z(6, 8) = 72, \quad Z(0, 5) = 30$

2 Pick the smallest value.
$Z_{\min} = 12$ at both $(0, 2)$ and $(3, 0)$

3 Two adjacent corners give the same minimum, so every point between them gives $Z = 12$. ✓ 1
Both corners lie on the line $4x + 6y = 12$.

4 Hence the minimum occurs at any point on the segment joining $(0, 2)$ and $(3, 0)$; option (D).

Answer **(D) any point on the line segment joining $(0, 2)$ and $(3, 0)$**

Watch out: Choosing (A) or (B): when two corners tie for the minimum, the whole edge between them is optimal.

2 MCQ 1 mark

The region represented by the system of inequations $3x + y \geq 3,\ 2x - y \geq -5,\ x, y \geq 0$ is:

(A) **unbounded in the 1st quadrant** **(B)** bounded in the 1st quadrant

(C) unbounded in the 2nd quadrant **(D)** bounded in the 2nd quadrant

Formula / idea: A feasible region is unbounded if it cannot be enclosed in a circle; $x, y \geq 0$ keeps it in the 1st quadrant.

1 Rewrite the second inequation.

$2x - y \geq -5 \ \Rightarrow \ y \leq 2x + 5$

2 Points on the boundary lines.

$3x + y = 3 : (1, 0),\ (0, 3)$

$y = 2x + 5 : (0, 5),\ (1, 7)$

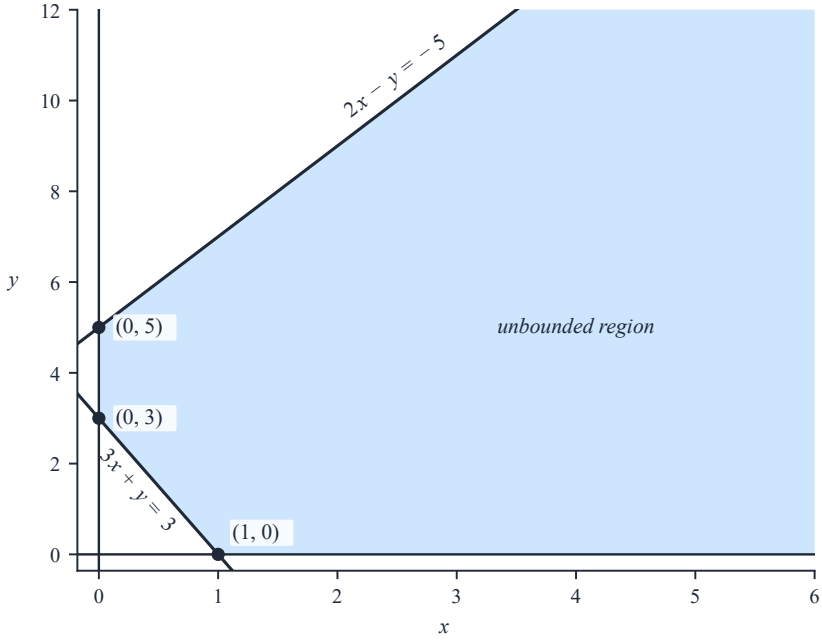
3 Origin test: $(0, 0)$ fails $3x + y \geq 3$ but satisfies $y \leq 2x + 5$.

$0 \geq 3\ (\text{false}), \qquad 0 \leq 5\ (\text{true})$

4 Since $x \geq 0,\ y \geq 0$, the region lies in the 1st quadrant only.

5 Every point $(k, 0)$ with $k \geq 1$ satisfies all the inequations, so the region extends without end. ✓ 1

$3k + 0 \geq 3, \qquad 2k - 0 \geq -5 \qquad (k \geq 1)$



6 Hence the region is unbounded in the 1st quadrant; option (A).

Answer **(A) unbounded in the 1st quadrant**

Watch out: Thinking the region is bounded because both lines cut the y-axis; it is open towards the right.

3 MCQ 1 mark

The restrictions imposed on the decision variables involved in the objective function of a linear programming problem are called:

(A) feasible solutions **(B)** **constraints**

(C) optimal solutions **(D)** infeasible solutions

1	Recall: the linear inequations or equations on the decision variables of an LPP are called **constraints**.	✓ 1
2	Feasible, optimal and infeasible solutions are points, not restrictions.	
3	Hence the answer is option (B) constraints.	

Answer **(B) constraints**

4 Assertion-Reason 1 mark

Assertion (A): If the feasible region of an LPP is bounded, the objective function $Z = ax + by$ has both a maximum and a minimum value on it. **Reason (R):** If the feasible region is bounded, the maximum and minimum values of Z each occur at a corner point of the feasible region.

(A) **Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).**

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

Formula / idea: Theorem 2 (NCERT): on a bounded feasible region, Z has both a maximum and a minimum, each at a corner point.

1	Assertion (A) is true: it is the first half of Theorem 2.	
2	Reason (R) is true: it is the second half of Theorem 2.	
3	A bounded region has finitely many corner points, so Z takes finitely many values there. The largest and smallest of these are the maximum and minimum.	
4	So R tells us why the maximum and minimum exist: R correctly explains A.	✓ 1
5	Hence the correct option is (A).	

Answer **(A) Both A and R are true, and R is the correct explanation of A**

Level 2

9 marks

5 **Short answer** 3 marks

Solve the following linear programming problem graphically: Minimise $Z = 600x + 400y$ subject to the constraints $x + y \geq 8$, $x + 2y \leq 16$, $4x + y \leq 29$, $x, y \geq 0$.

Formula / idea: Corner Point Method: on a bounded region, evaluate Z at every corner point and pick the least value.

1 Treat each constraint as an equation and find two points on each line.

$x + y = 8 : (8, 0),\ (0, 8)$

$x + 2y = 16 : (16, 0),\ (0, 8)$

$4x + y = 29 : (7.25, 0),\ (0, 29)$

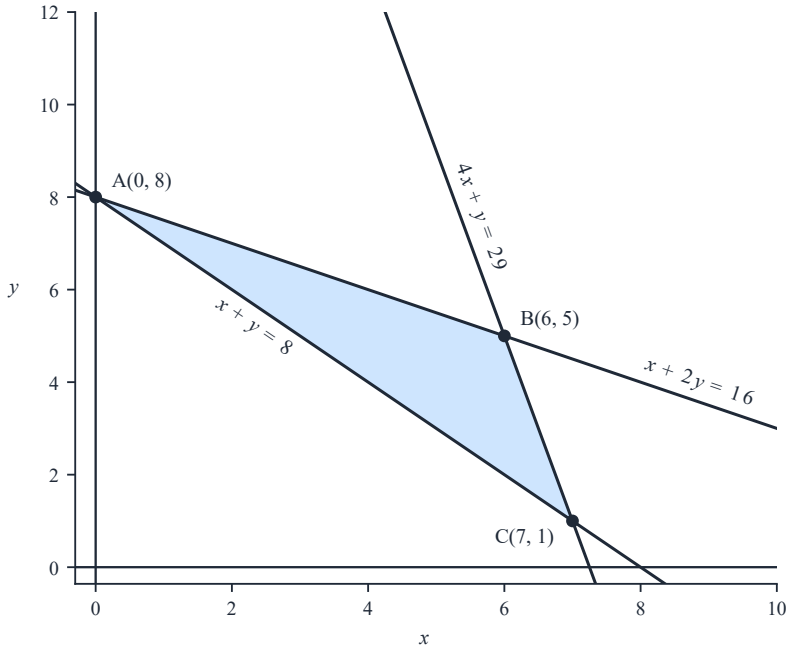
2 Origin test: $(0, 0)$ fails $x + y \geq 8$, so shade away from the origin.

$0 + 0 \geq 8\ (\text{false})$

3 $(0, 0)$ satisfies $x + 2y \leq 16$ and $4x + y \leq 29$, so shade towards the origin.

$0 \leq 16, \quad 0 \leq 29\ (\text{true})$

4 Draw the lines and shade the common region in the 1st quadrant: bounded triangle ABC. ✓ 1½



5 Corner B: solve $x + 2y = 16$ and $4x + y = 29$.

$x = 16 - 2y \Rightarrow 4(16 - 2y) + y = 29 \Rightarrow 7y = 35 \Rightarrow y = 5,\ x = 6$

6 Corner C: subtract $x + y = 8$ from $4x + y = 29$.

$3x = 21 \Rightarrow x = 7,\ y = 1$

7 Corner A: both $x + y = 8$ and $x + 2y = 16$ pass through $(0, 8)$.

$A(0, 8), \quad B(6, 5), \quad C(7, 1)$

8 Find Z at each corner point. ✓ 1

Corner point	$Z = 600x + 400y$
$A(0, 8)$	$3200 \leftarrow$ minimum
$B(6, 5)$	5600
$C(7, 1)$	4600

9 The region is bounded, so the least corner value is the minimum. ✓ ½

$Z_{\min} = 3200$ at $x = 0,\ y = 8$

Answer **Minimum** $Z = 3200$ **at** $x = 0,\ y = 8$

Watch out: Shading $x + y \geq 8$ towards the origin; always run the origin test for each constraint.

6 Short answer — 3 marks

Solve the following LPP graphically: Minimise $Z = 3x + 5y$ subject to $x + 3y \geq 3$, $x + y \geq 2$, $x \geq 0$, $y \geq 0$.

Formula / idea: On an unbounded region, the smallest corner value m is the minimum only if $ax + by < m$ has no point in common with the region.

1 Find two points on each boundary line.

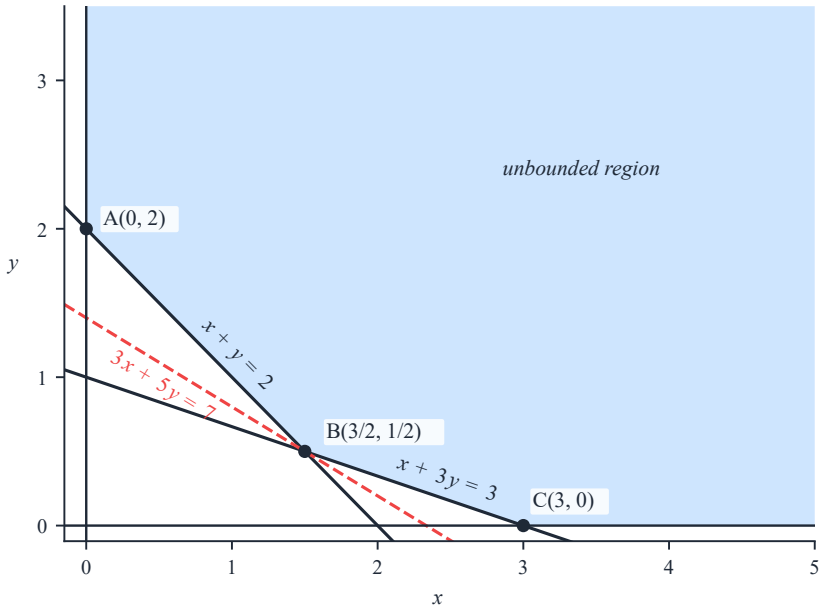
$x + 3y = 3 : (3, 0),\ (0, 1)$

$x + y = 2 : (2, 0),\ (0, 2)$

2 Origin test: $(0, 0)$ fails both constraints, so shade away from the origin.

$0 \geq 3$ (false), $\quad 0 \geq 2$ (false)

3 Draw the lines and shade the common region in the 1st quadrant; it is unbounded. ✓ 1½



4 Corner B: subtract $x + y = 2$ from $x + 3y = 3$.

$2y = 1 \Rightarrow y = \frac{1}{2}, \quad x = 2 - \frac{1}{2} = \frac{3}{2}$

5 Find Z at each corner point. ✓ 1

Corner point	$Z = 3x + 5y$
$A(0, 2)$	10
$B\left(\frac{3}{2}, \frac{1}{2}\right)$	$7 \leftarrow$ smallest
$C(3, 0)$	9

6 The region is unbounded, so 7 may or may not be the minimum.

An unbounded region needs the half-plane check.

7 Draw the open half-plane $3x + 5y < 7$ (below the dashed line through B).

$3x + 5y < 7$

8 It has no point in common with the feasible region.

9 Hence the minimum value of Z is 7. ✓ ½

$Z_{\min} = 7$ at $x = \frac{3}{2}, \ y = \frac{1}{2}$

Answer **Minimum $Z = 7$ at $x = \frac{3}{2}, \ y = \frac{1}{2}$**

Watch out: Writing the minimum straight from the table; for an unbounded region the half-plane check must be shown.

7 **Short answer** 3 marks

Solve the following LPP graphically: Maximise $Z = 5x + 3y$ subject to $3x + 5y \leq 15$, $5x + 2y \leq 10$, $x \geq 0, y \geq 0$.

Formula / idea: Corner Point Method: on a bounded region, evaluate Z at every corner point and pick the greatest value.

1 Find two points on each boundary line.

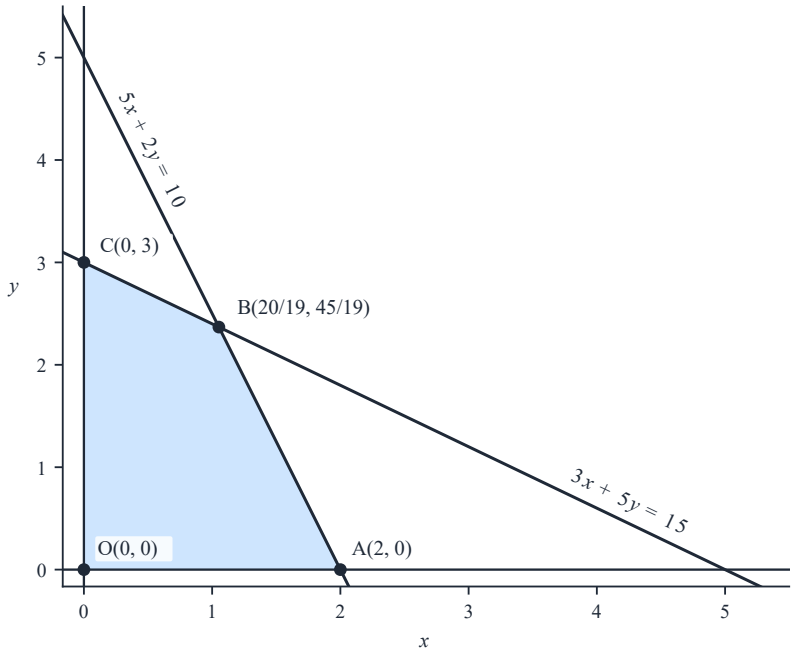
$3x + 5y = 15 : (5, 0), \ (0, 3)$

$5x + 2y = 10 : (2, 0), \ (0, 5)$

2 Origin test: $(0, 0)$ satisfies both constraints, so shade towards the origin.

$0 \leq 15, \quad 0 \leq 10 \ (\text{true})$

3 Draw the lines and shade the common region in the 1st quadrant: bounded region $OABC$. ✓ 1½



4 Corner B: multiply to equalise the y-coefficients, then subtract.

$2(3x + 5y) = 30 \Rightarrow 6x + 10y = 30$

$5(5x + 2y) = 50 \Rightarrow 25x + 10y = 50$

$\Rightarrow 19x = 20 \Rightarrow x = \dfrac{20}{19}$

5 Put $x = \dfrac{20}{19}$ in $5x + 2y = 10$.

$2y = 10 - \dfrac{100}{19} = \dfrac{90}{19} \Rightarrow y = \dfrac{45}{19}$

6 Find Z at each corner point. ✓ 1

Corner point	$Z = 5x + 3y$
$O(0, 0)$	0
$A(2, 0)$	10
$B\left(\frac{20}{19}, \frac{45}{19}\right)$	$\frac{100}{19} + \frac{135}{19} = \frac{235}{19}$ ← maximum
$C(0, 3)$	9

7 The region is bounded, so the greatest corner value is the maximum. ✓ ½

$Z_{\max} = \dfrac{235}{19} \approx 12.37$ at $x = \dfrac{20}{19}$, $y = \dfrac{45}{19}$

Answer **Maximum** $Z = \dfrac{235}{19}$ **at** $x = \dfrac{20}{19}$, $y = \dfrac{45}{19}$

Watch out: Skipping the intersection point B; the maximum is at B, not at the axis corners.

Level 3

14 marks

8 **Long answer** 5 marks

Solve graphically: Minimise and maximise $Z = x + 2y$ subject to $x + 2y \geq 100$, $2x - y \leq 0$, $2x + y \leq 200$, $x \geq 0$, $y \geq 0$.

Formula / idea: Corner Point Method; if two adjacent corners give the same optimum, every point of the edge joining them is optimal.

1 Find two points on each boundary line.

$x + 2y = 100 : (100, 0),\ (0, 50)$

$2x - y = 0 : (0, 0),\ (50, 100)$

$2x + y = 200 : (100, 0),\ (0, 200)$

2 Origin test: $(0, 0)$ fails $x + 2y \geq 100$ (shade away) and satisfies $2x + y \leq 200$ (shade towards).

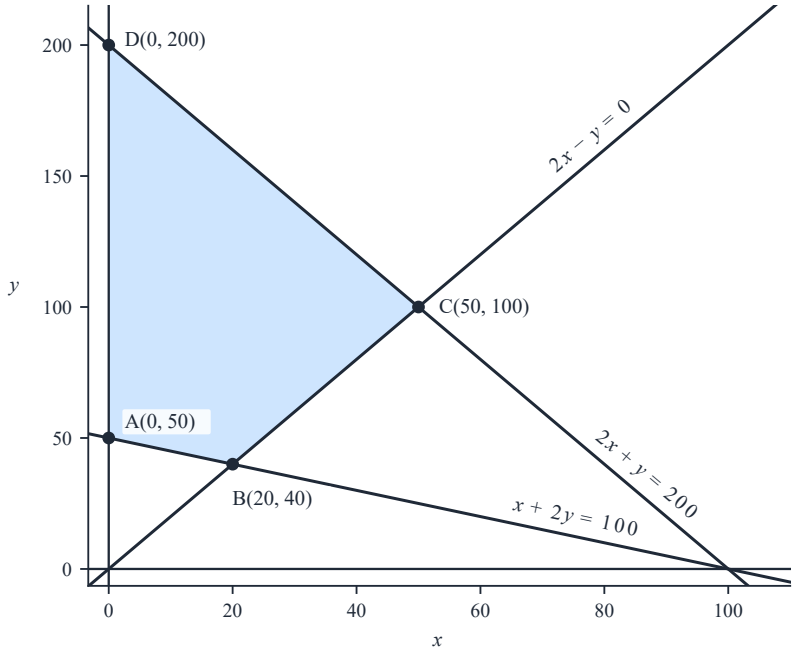
$0 \geq 100\ (\text{false}), \quad 0 \leq 200\ (\text{true})$

3 $2x - y = 0$ passes through the origin, so test $(0, 1)$ instead.

$2(0) - 1 = -1 \leq 0\ (\text{true})$

So shade the side above the line $y = 2x$.

4 Draw the lines and shade the common region in the 1st quadrant: bounded region $ABCD$. ✓ 2



5 Corner B: put $y = 2x$ in $x + 2y = 100$. ✓ ½

$x + 4x = 100 \Rightarrow x = 20,\ y = 40$

6 Corner C: put $y = 2x$ in $2x + y = 200$. ✓ ½

$2x + 2x = 200 \Rightarrow x = 50,\ y = 100$

7 Corners on the y-axis come from the intercepts.
$A(0, 50), \quad B(20, 40), \quad C(50, 100), \quad D(0, 200)$

8 Find Z at each corner point. ✓ 1

Corner point	$Z = x + 2y$
$A(0, 50)$	$100 \leftarrow$ minimum
$B(20, 40)$	$100 \leftarrow$ minimum
$C(50, 100)$	250
$D(0, 200)$	$400 \leftarrow$ maximum

9 The region is bounded; the minimum 100 occurs at both A and B. ✓ ½
$Z_{\min} = 100$ at all points on the segment joining $(0, 50)$ and $(20, 40)$
AB lies on $x + 2y = 100$, so $Z = 100$ along it.

10 The maximum is the greatest corner value. ✓ ½
$Z_{\max} = 400$ at $(0, 200)$

Answer **Minimum $Z = 100$ at every point of the segment joining $(0, 50)$ and $(20, 40)$; maximum $Z = 400$ at $(0, 200)$**

Watch out: Writing the minimum at only one point; A and B tie, so the whole segment AB gives the minimum.

9 Long answer — 5 marks

A furniture dealer deals in only two items, tables and chairs. He has ₹50,000 to invest and storage space for at most 60 pieces. A table costs ₹2,500 and a chair ₹500. He expects a profit of ₹250 on a table and ₹75 on a chair. Formulate this as an LPP and solve it graphically to find how many tables and chairs he should buy to maximise his profit, assuming he can sell all the items he buys.

Formula / idea: Formulate (variables, objective, constraints), graph the feasible region, then use the Corner Point Method.

1 Let the dealer buy x tables and y chairs. ✓ ½

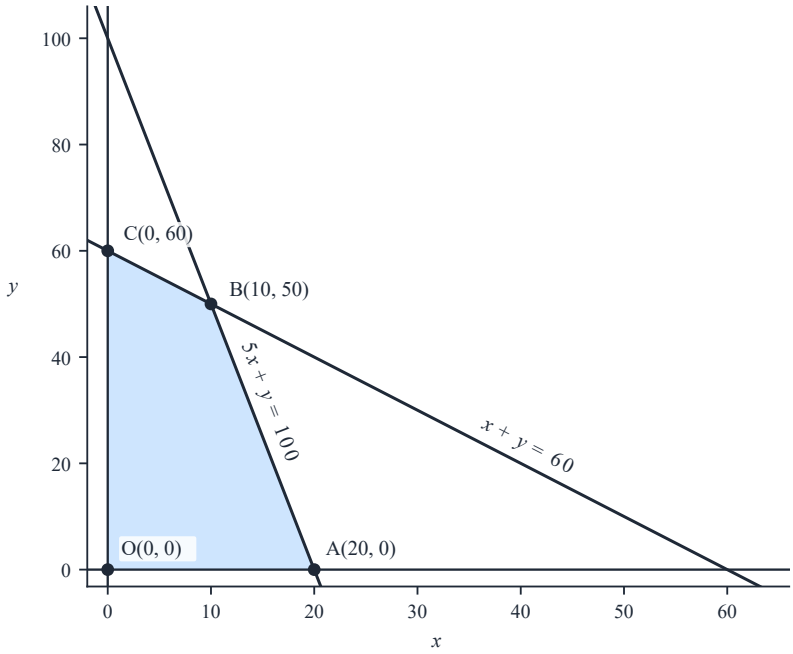
2 Profit is ₹250 per table and ₹75 per chair. ✓ ½
$\text{Maximise } Z = 250x + 75y$

3 Investment constraint (₹2,500 per table, ₹500 per chair, ₹50,000 in all); divide by 500. ✓ ½
$2500x + 500y \leq 50000 \;\Rightarrow\; 5x + y \leq 100$

4 Storage constraint (at most 60 pieces) and non-negativity. ✓ ½
$x + y \leq 60, \qquad x \geq 0,\ y \geq 0$

5 Find two points on each line; $(0, 0)$ satisfies both, so shade towards the origin.
$5x + y = 100 : (20, 0),\ (0, 100)$
$x + y = 60 : (60, 0),\ (0, 60)$

6 Draw the lines and shade the common region in the 1st quadrant: bounded region $OABC$. ✓ 1½



7 Corner B: subtract $x + y = 60$ from $5x + y = 100$.

$4x = 40 \Rightarrow x = 10, \quad y = 60 - 10 = 50$

8 Find Z at each corner point. ✓ 1

Corner point	$Z = 250x + 75y$
$O(0, 0)$	0
$A(20, 0)$	5000
$B(10, 50)$	$6250 \leftarrow$ maximum
$C(0, 60)$	4500

9 The region is bounded, so the maximum is at $B(10, 50)$.

$Z_{\max} = 6250$ at $x = 10,\ y = 50$

10 Hence the dealer should buy 10 tables and 50 chairs; maximum profit is ₹6,250. ✓ ½

Answer **Buy 10 tables and 50 chairs; maximum profit ₹6,250**

Watch out: Not dividing $2500x + 500y \leq 50000$ by 500, which makes the graph scale awkward and errors likely.

10 **Case study** 4 marks

Case Study: A dietician wants to prepare a mixture of two foods F_1 and F_2. One kg of F_1 contains 2 units of vitamin A and 1 unit of minerals; one kg of F_2 contains 1 unit of vitamin A and 2 units of minerals. The mixture must contain at least 8 units of vitamin A and 10 units of minerals. F_1 costs ₹50 per kg and F_2 costs ₹70 per kg. Let x kg of F_1 and y kg of F_2 be used.

(i) Write the objective function and the constraints of the LPP. 1 mark

(ii) Find the corner points of the feasible region. 1 mark

(iii) Find the minimum cost of the mixture, and justify why the minimum exists even though the feasible region is unbounded. 2 marks

Formula / idea: On an unbounded region, the smallest corner value m is the minimum only if $ax + by < m$ has no point in common with the region.

(i) 1 mark

1 Cost is ₹50 per kg of F_1 and ₹70 per kg of F_2.
$\text{Minimise } Z = 50x + 70y$ ✓ ½

2 Write the vitamin A, mineral and non-negativity constraints.
$2x + y \geq 8 \quad (\text{vitamin A})$
$x + 2y \geq 10 \quad (\text{minerals})$
$x \geq 0,\ y \geq 0$ ✓ ½

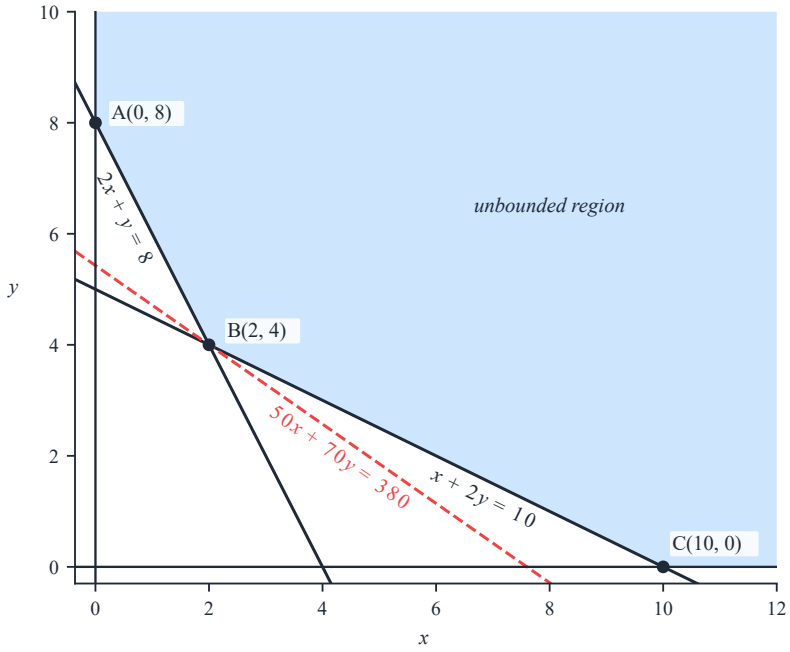
Answer (i) **Minimise** $Z = 50x + 70y$ **subject to** $2x + y \geq 8$, $x + 2y \geq 10$, $x \geq 0$, $y \geq 0$

(ii) 1 mark

1 Find two points on each boundary line.
$2x + y = 8 : (4, 0),\ (0, 8)$
$x + 2y = 10 : (10, 0),\ (0, 5)$

2 $(0, 0)$ fails both constraints, so shade away from the origin; the region is unbounded. ✓ ½

$0 \geq 8$ (false), $0 \geq 10$ (false)



3 Corner B: put $y = 8 - 2x$ in $x + 2y = 10$.

$x + 16 - 4x = 10 \Rightarrow x = 2, \quad y = 8 - 4 = 4$

4 List the corner points. ✓ ½

$A(0, 8), \quad B(2, 4), \quad C(10, 0)$

Answer (ii) $A(0, 8)$, $B(2, 4)$, $C(10, 0)$

(iii)

2 marks

1 Find Z at each corner point. ✓ 1

Corner point	$Z = 50x + 70y$
$A(0, 8)$	560
$B(2, 4)$	380 ← smallest
$C(10, 0)$	500

2 The region is unbounded, so 380 may or may not be the minimum.

3 Draw the open half-plane $50x + 70y < 380$, i.e. $5x + 7y < 38$ (dashed line).

$50x + 70y < 380$

4 It has no point in common with the feasible region, so 380 is the minimum. ✓ ½

5 Hence the minimum cost is ₹380, with 2 kg of F_1 and 4 kg of F_2. ✓ ½

$Z_{\min} = 380$ at $x = 2, \ y = 4$

Answer (iii) **Minimum cost ₹380 at $x = 2$, $y = 4$; it exists because $50x + 70y < 380$ has no point in common with the region**

Watch out: Writing $\leq$ for "at least"; "at least 8 units" means $2x + y \geq 8$.